

1. What is one key advantage of using domain-specific libraries like TorchVision for vision tasks? 1 point
- ☐ They provide generic tools applicable to any data type.
 - ☐ They eliminate the need for training models.
 - ☒ They offer optimized tools, datasets, and pretrained models tailored for vision workflows.
 - ☐ They automatically improve accuracy of models.
2. Which library does TorchVision rely on for image file I/O and manipulation before tensor conversion? 1 point
- ☐ NumPy
 - ☐ OpenCV
 - ☒ PIL (Pillow)
 - ☐ seaborn
3. What is the primary goal of transfer learning? 1 point
- ☒ To leverage knowledge gained from pre-training on large datasets.
 - ☐ To train models from scratch
 - ☐ To combine multiple model architectures into a single ensemble for better prediction.
 - ☐ To reduce the computational cost of inference.
4. What will the following code do? 1 point
- ```
boxes = torch.tensor([[140, 30, 375, 315], [200, 70, 230, 110]])

draw_bounding_boxes(image, boxes, labels=["dog", "eye"], colors=["red", "blue"])
```
- ☐ It saves the bounding boxes to disk.
  - ☒ It returns a new image tensor with boxes and labels drawn.
  - ☐ It modifies the original image in place.
  - ☐ It runs object detection and returns the predicted boxes.
5. Why would you apply RandomResizedCrop() instead of just RandomCrop() in a training pipeline for image classification? 1 point
- ☐ To crop images at a random location but with a fixed aspect ratio
  - ☐ To introduce random cropping along with random resizing, simulating scale variation
  - ☒ To ensure all crops are center-aligned after resizing
  - ☐ To normalize pixel intensity ranges after cropping
6. In an image classification pipeline, why is it important to apply Resize() before CenterCrop() or RandomCrop()? 1 point
- ☐ To convert the image to grayscale
  - ☐ To change the image color depth
  - ☒ To ensure the image is large enough to safely crop to the target size
  - ☐ To remove background artifacts before cropping
7. In a well-designed image pipeline, where should Normalize() typically be applied? 1 point
- ☐ Before loading the image from disk
  - ☐ Before converting the image to a tensor
  - ☒ Immediately after converting to a tensor with ToTensor()
  - ☐ After training is complete
8. What output does a pretrained DeepLabV3 model from torchvision.models.segmentation produce? 1 point
- ☐ A list of bounding boxes
  - ☐ A tensor of class scores for each image
  - ☒ A per-pixel tensor indicating predicted class for each pixel
  - ☐ A prediction on what class is found in an image
9. Fill in the missing line to correctly preprocess the image for a pretrained segmentation model: 1 point
- ```
img = Image.open("dog.jpg")  
  
transform = transforms.Compose([  
    transforms.ToTensor(),  
    _____  
])  
  
input_tensor = transform(img).unsqueeze(0)
```
- ☐ transforms.CenterCrop(224)
 - ☐ transforms.Grayscale()
 - ☒ transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])
 - ☐ transforms.Resize((32, 32))

10. Why is `unsqueeze(0)` often used after converting an image to a tensor for model input?

1 point

- ☐ It adds a color channel to grayscale images
- ☐ It ensures the pixel values are between 0 and 1
- ☒ It adds a batch dimension so the input shape is [1, C, H, W]
- ☐ It flattens the image for matrix multiplication